

Jiayi (Marco) Chen

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Research Interests: Information & Mechanism Design, Decision-Making under Uncertainty, Multi-agent System.

EDUCATION

- **Columbia University** New York, NY
M.A. in Statistics Sep 2025 - Dec 2026
 - Doctoral-Level Coursework: Convex Optimization, Stochastic Modeling, Reinforcement Learning, Diffusion Models
 - Additional Coursework: Modern Analysis, Honors Probability Theory/Linear Regression/ Statistical Inference
- **Soochow University** Suzhou, China
B.A. in Accounting, Minor in Applied Psychology; GPA: 3.7/4.0 Sep 2021 - Jun 2025
 - Selected Coursework: Calculus, Linear Algebra, Probability & Statistics
- **University of California, Berkeley** Berkeley, CA
Visiting Student, Berkeley Global Access Program (IEOR); GPA: 3.95/4.00 Jan 2024 - May 2024
 - Selected Coursework: Introduction to Optimization Modeling, Introduction to Stochastic Processes, Machine Learning and Data Analytics II, Analysis and Design of Databases

WORKING PAPERS

- **Information Design for Mitigating Volunteer's Dilemma**, with Weiliang Liu, Tao Lin, and Wei Tang.

RESEARCH EXPERIENCE

- **Information Design for Mitigating Volunteer's Dilemma** Remote
Research Assistant; Advisors: Prof. Weiliang Liu, Tao Lin, Wei Tang Jul 2025 - Present
 - Formulated a volunteer dispatch platform's disclosure of responder availability as Bayesian persuasion with multiple strategic receivers, maximizing the probability of at least one response under symmetric equilibrium
 - Proved a support-reduction theorem for arbitrary population sizes, reducing the nontrivial disclosure problem to a finite-dimensional convexification over two posterior regions
 - Established a sharp uniform 3/4-approximation guarantee for full information relative to the optimal provision probability; extended the bound to private costs stochastically dominated by the uniform benchmark
 - Calibrated the model to published statistics from 5,646 Dutch cardiac-arrest dispatch activations; benchmarked policies using grid-based linear programming with local refinement, finding full information achieved 99.05% of the computed optimum in the baseline calibration
- **Coarse Understanding in Persuasion** New York, NY
Research Assistant; Advisor: Prof. Yuchen Hu Jun 2026 - Present
 - Developing a model of feature highlighting in which receivers understand selection policies through coarse state partitions, nesting naive and fully Bayesian inference
 - Characterizing which policy-relevant distinctions suffice for Bayesian inference and how ignored within-cell selection variation affects posterior predictions
- **Framework-agnostic Evaluation for Healthcare Agents** Remote
Research Assistant; Advisor: Prof. Yu Ma May 2026 - Present
 - Developing a framework-agnostic evaluation protocol for healthcare multi-agent systems that distinguishes candidate availability, decision errors, and execution failures
 - Designing paired interventions to evaluate communication, refinement, and routing, and extending evaluation from medical question answering to EHR tool-use tasks
- **Eating Context and Well-Being: A Daily Diary Study** Suzhou, China
Undergraduate Thesis; Advisor: Prof. Chujun Wang Dec 2024 - Apr 2025
 - Analyzed 420 daily records from 60 participants in a 7-day diary study using linear mixed models (REML) and Bayesian mixed models in R to examine associations between eating context, meal calorie ratings, and emotional well-being
 - Estimated asymmetric within-person changes in photo-based meal calorie ratings: +29.1% for solitary-to-communal transitions (95% credible interval: 8.5% to 48.9%), versus -3.8% in the reverse direction, with the latter interval spanning zero

PRESENTATIONS

- **Information Design for Mitigating Volunteer's Dilemma** San Francisco, CA
Invited session (INFORMS): Data-Driven Optimization and Analytics for Social Impact Nov 2026

SELECTED PROJECTS

- **Constrained Reinforcement Learning for EMS Dispatch** Columbia University
Course Project, ORCS 6529 Advanced Reinforcement Learning Spring 2026
 - Developed RG-CPO, a PPO-Lagrangian method with PID dual control and CVaR constraints, for constrained ambulance dispatch policies
 - Calibrated and evaluated the policy on NYC EMS data, then presented the results in class
- **Course Scheduling via Mixed Integer Programming** UC Berkeley
Course Project, IEOR 164 Optimization Modeling Apr 2024 - May 2024
 - Formulated and solved a MIP in Python/Pyomo for classroom allocation and course timing, integrating capacity, timing, and inter-building distance constraints calibrated on historical enrollment data
 - Incorporated soft penalties for early-morning slots, lunchtime conflicts, and consecutive classes in distant buildings, producing a conflict-free schedule
- **Brain Tumor Detection via Machine Learning** UC Berkeley
Course Project, IEOR 142B Machine Learning and Data Analytics II Mar 2024 - May 2024
 - Trained and benchmarked object detection models (RetinaNet, Faster R-CNN) for tumor identification in MRI images; the non-augmented RetinaNet dominated at fixed IoU thresholds (AP 0.109, AR 0.700)

INDUSTRY EXPERIENCE

- **Deloitte Touche Tohmatsu Co., Ltd.** Shanghai, China
Data Analyst Intern, iBond Oct 2024 - Nov 2024
 - Built a bond default risk scoring model integrating 53+ indicators across sentiment, fundamentals, volume-price, and issuance-rate dimensions (85%+ out-of-sample accuracy)
 - Designed a 10-tier early warning system based on historical percentiles, enabling preemptive flagging of bonds with default probabilities above 90%
- **Bank of China Investment Management Co., Ltd.** Shanghai, China
Quantitative Research Intern, Overseas and Quantitative Index Department Jul 2024 - Aug 2024
 - Developed factor models for Fund of Funds selection over roughly 120 million records of returns and holdings data, reaching an Information Coefficient of 0.1
 - Improved volume-price factors through genetic programming to reduce cross-factor correlation and applied KAN-based nonlinear modeling, achieving 6% annualized excess return against the CSI 500 benchmark

SKILLS SUMMARY

- **Languages:** Python (PyTorch, Pyomo, Gurobi, NumPy, Pandas, scikit-learn), R, SQL, L^AT_EX, Tableau
- **Spoken:** Mandarin (native), English (fluent), Japanese
- **Interests:** Chess, Calligraphy, Classical Guitar